

Number theory – 4

p -adic valuations, factorials and binomial coefficients

Homework

14 сентября 2026 года

Theoretical minimum. The valuation $v_p(n)$ is the exponent of p in n ; it is additive on products and satisfies $v_p(x+y) \geq \min(v_p(x), v_p(y))$ with equality when the valuations differ. Legendre's formula gives $v_p(n!) = \sum_{i \geq 1} \lfloor n/p^i \rfloor = (n - s_p(n))/(p-1)$ with s_p the digit sum in base p . Kummer's theorem says $v_p\binom{n}{k}$ is the number of carries when k and $n-k$ are added in base p ; hence all $\binom{n}{k}$ are odd exactly for $n = 2^m - 1$, and $\binom{2n}{n}$ has no prime factor between $2n/3$ and n . Lifting the exponent: for odd p with $p \mid a-b$ and $p \nmid a$, $v_p(a^n - b^n) = v_p(a-b) + v_p(n)$; for $p=2$ and even n there is the extra term $v_2(a+b) - 1$. Lucas' theorem is in Combinatorics – 3, Stirling's formula in Limits – 6, Chebyshev's bounds in Series and infinite products – 6.

1. Prove Erdős's estimate $\prod_{p \leq n} p < 4^n$ by induction using $\binom{2m+1}{m}$, and combine it with Kummer's theorem to prove Bertrand's postulate: for every $n \geq 1$ there is a prime between n and $2n$.
2. Prove that $\text{lcm}(1, 2, \dots, n) = e^{(1+o(1))n}$ and that $\binom{2n}{n}$ is divisible by $\text{lcm}(1, \dots, n+1)/(n+1)$; deduce a lower bound for the number of primes up to $2n$.
3. Determine all positive integers n such that $v_2\left(\binom{2n}{n}\right) = 1$, and describe the set of n for which $\binom{2n}{n}$ is squarefree.
4. Let p be an odd prime. Prove that $v_p\left(\sum_{k=1}^{p-1} \frac{1}{k}\right) \geq 2$ for $p \geq 5$ and compute $v_p\left(\sum_{k=1}^{p-1} \frac{1}{k^2}\right)$.
5. Let $a_k = \lfloor n^k/k \rfloor$ for a fixed integer $n > 1$. Prove that infinitely many terms of the sequence $(a_k)_{k \geq 1}$ are odd. (IMO shortlist 2014, N4)

6. For each positive integer k let $t(k)$ be the largest odd divisor of k . Determine all positive integers a for which there exists a positive integer n such that all the differences $t(n+a) - t(n)$, $t(n+a+1) - t(n+1)$, $\dots$, $t(n+2a-1) - t(n+a-1)$ are divisible by 4. (IMO shortlist 2011, N4)
7. For $d > 0$ let $f(d)$ be the smallest positive integer with exactly d divisors. Prove that $f(2^{k+1})$ is divisible by $f(2^k)$ for every k . (IMO shortlist 2011, N1)
8. Prove that $\binom{2n}{n}$ is never a perfect power for $n > 1$, using Bertrand's postulate and Kummer's theorem.
9. Prove that for every n the number $\text{lcm}(\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n})$ equals $\text{lcm}(1, 2, \dots, n+1)/(n+1)$, and deduce the identity $\sum_k \frac{(-1)^k}{\binom{n}{k}}$ in closed form.
10. Let p be a prime and let n be written in base p . Prove that the exponent of p in the multinomial coefficient $\frac{n!}{k_1! \dots k_m!}$ equals the number of carries in the base- p addition of $k_1, \dots, k_m$, and use it to determine when a multinomial coefficient is coprime to p .